LE NHAT KHA (KAI)

AI Research Engineer

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SUMMARY

AI Research Engineer with 2+ years of experience in applied research and product development within technology companies. Expertise spans multiple domains, including Computer Vision (OCR, eKYC, deepfake detection, generative models), Multimodal Large Language Models, and Speech Processing. Demonstrated ability to translate research into production-grade systems. Published one peer-reviewed paper at WACV.

SKILLS

Technology: Git, Docker, ComfyUI, PyTorch, TensorFlow, Scikit-learn, Gradio, Hugging Face, Triton Inference Server, vLLM, SGLang

Knowledge: Machine Learning, Deep Learning, Reinforcement Learning, Computer Vision, Natural Language Processing, Maths, LLM/VLM, Traffic Simulation (SUMO)

EDUCATION

11/2020 - 5/2024 **Viet Nam National University Ho Chi Minh City - VNUHCM (UIT)** **Bachelor of Science - BS**

Major: Computer Science, **Honors program**. Highest CGPA in Faculty of CS (**3.87/4.00**)

Thesis: *Gradual Domain Adaptation in Scene Text Recognition* (mark: **9.9/10**)

Research member at **CORE Lab** — Computer Vision & Retrieval

5x Recipient of Scholarships for Excellent Academic Record (**100%** tuition coverage per semester)

9/2017 - 5/2020 **Le Quy Don High School for the Gifted, Da Nang** **High School**

Specialised-subject: Physics. GPA (grade 12): **9.6/10** (ranked **1st** in the class)

Executive Committee Member, Green Season Journalism Club

PUBLICATIONS

2025 **Le, Kha Nhat**, Hoang-Tuan Nguyen, Hung Tien Tran, and Thanh Duc Ngo. "Stratified Domain Adaptation: A Progressive Self-Training Approach for Scene Text Recognition." In *2025 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)* (pp. 8990–9000). IEEE, 2025.

PROFESSIONAL EXPERIENCE

8/2026 – *present* **AI Engineer** **Agntic.ai**

- *About company*: Agntic provides a conversational platform for businesses to recruit and manage AI employees, enabling rapid deployment without coding or specialized AI expertise.
- *Job description*: Develop and integrate AI agents and LLM-powered systems into the Agntic.ai platform, focusing on agent workflows, tool use, automation, and reliable production deployment.

5/2026 – 7/2026 **AI Research Assistant** **VinUniversity (CAIR)**

- *About company*: The Center for AI Research (CAIR) at VinUniversity is dedicated to advancing the next generation of intelligence across brain science, AI, and robotics.
- *Job description*: Modeling and simulation of traffic in Ho Chi Minh City, as part of the project *Adaptive Reinforcement Learning for Scalable Traffic Control under Uncertainty*, under the supervision of Dr Duc Thinh Pham (Chief Scientist at Air Traffic Management Research Institute - NTU).

8/2024 – 4/2026 **AI Research Engineer** **MoMo (M_service)**

- *About company*: MoMo is a leading super app and one of Vietnam's fastest-growing fintech companies, boasting over 31 million users and more than 50,000 domestic partners.
- *Job description*: Integrate AI and Machine Learning as core components across various product functions, including Vision-Language Models (VLMs), Generative AI (image), eKYC (FlashMark), and fraud detection.

4/2024 – 7/2024 **Viettel Digital Talent 2024 (AI resident)** **Viettel AI**

- *About company*: Viettel, the largest telecommunications group in Vietnam with 76 million customers, offers the Viettel Digital Talent program—an intensive and highly competitive training initiative with over 3,000 applicants (acceptance rate: 9%).
- *Job description*: Developed an innovative approach to enhance Non-Intrusive Speech Quality Assessment (NISQA) models by integrating the CMGAN model from the Speech Enhancement task

6/2023 – 8/2023 **Cinnamon AI Bootcamp 2023 (trainee)** **Cinnamon AI**

- *About company*: Cinnamon AI is the pioneer in business consulting and developing AI solutions. The Cinnamon AI Bootcamp is an annual immersive program for students, combining research and theory with an engineering mindset and practical, hands-on experience.
- *Job description*: Developed a 3D motion-capturing system, which involved using $\mathrm{E}^2\mathrm{FGVI}$ for video inpainting, and FrankMocap for pose estimation and alignment of a mannequin to the scene

PROJECTS

4/2025 – 5/2026 **MoMo VL – Large Vision-Language Model** **MoMo (M_service)**

- *Description*:
  - Developed a Large Vision-Language Model (LVLM) to address key user-facing tasks at MoMo, including Expense Management, Merchant Validation, MoMo Local Guide, and text extraction tasks
- *Contributions*:
  - Managed a team of 2 collaborators responsible for crawling and filtering data from Vietnamese websites such as CafeF, VnExpress, SachVui, and CuuDuongThanCong
  - Finetuned open-source models (e.g., Qwen2.5VL, InternVL3) on Vietnamese datasets, and applied reinforcement learning techniques (DPO and MPO) to enhance the models' reasoning capabilities

8/2024 – 5/2026 **Flashmark eKYC – Passive Liveness Detection System** **MoMo (M_service)**

- *Description*:
  - Developed a face-based user authentication system to enhance app security and prevent spoofing attacks (input: user's selfie image, output: real user or cheater?)
  - Flashmark is a cutting-edge passive liveness detection solution with high security standards. By flashing a sequence of colors on the screen and capturing frames where the user's face reflects the corresponding hues, the system can accurately determine whether the input is from a real user.
- *Contributions*:
  - Conducted research, built own dataset, designed and trained **4 AI modules** to comprehensively detect both physical attacks (photo spoofing, 3D mask attacks) and cyber attacks (deepfake, injection)
  - Managed a team of 4 collaborators responsible for collecting real-user data from the MoMo app, as well as creating spoofed data using deepfake and relighting tools
  - Contributed to enabling the "MoMo Face Authentication" system to achieve **iBeta Level 2** certification

2/2023 – 7/2024 **Stratified Domain Adaptation for Scene Text Recognition (StrDA)** KhaLee2307/StrDA

- *Description*:
  - Addressed the domain gap problem between the source - target domains (synthetic - real data) in scene text recognition (input: image; output: text extracted from the image)
  - This is the source code for my **first paper** with the first author role (WACV 2025 Early Acceptance)
- *Contributions*:
  - Main author, solution designer, experiment runner, and paper writer
  - Proposed **novel methods** to examine the gradual escalation of domain gap for the learning process. Extensive experiments show that our method achieves **state-of-the-art** results on the benchmark

AWARDS AND ACHIEVEMENTS

2020 **First Prize**, The Provincial Physics Competition

2019 **Second Prize**, The Provincial Science and Engineering Fair

2021 **Prize for Student of 5 merits**, University of Information Technology - VNUHCM

2025 **Certificate of Excellent Contribution**, Award from the CEO of MoMo

PROFESSIONAL DEVELOPMENT

8/2025 – 8/2025 **12th Vietnam Summer School of Science (VSSS - "Embrace the Unknown")** **Rencontres du Vietnam**
a multidisciplinary program focused on research methodology, academic integrity, scientific publishing, career development, and networking among early-career researchers

PROFESSIONAL SERVICE

Conference Reviewer/Program Committee: NeurIPS 2026

REFERENCES

Dr. Duc Thinh Pham
Principal Research Scientist and Chief Scientist of ATMRI, Nanyang Technological University
Assistant Professor College of Engineering and Computer Science, VinUniversity

Dr. Thanh Duc Ngo
Dean of Computer Science Faculty, University of Information Technology, VNU-HCM

Pham Kim Long
Director of AI & Big Data, Information Technology Center, MoMo (M_service)